

OTIC Studio

PRODUCT_REQUIREMENTS.MD

Version: 1.0 Owner: OTIC Technologies Status: Foundational Product Specification

PRODUCT VISION

OTIC Studio is an Offline AI Learning Operating System designed to help learners: Understand → Practice → Apply → Create → Collaborate → Teach Others The system must function completely offline. OTIC is: Educational App, AI Tutor, Learning Operating System, Education Platform (combined into one ecosystem).

MISSION

Enable anyone to learn anything, anywhere, without requiring internet access.

CORE PRINCIPLES

1) Learning must lead to action. Knowledge alone is insufficient. 2) Every answer must create understanding. 3) Every topic should become practical. 4) Learning should adapt to the learner. 5) OTIC should mentor, not merely answer.

USER ROLES

Guest — Purpose: Demonstration. Capabilities: Ask questions, View simulations, Explore platform. Limitations: No memory, No certificates, No progress tracking. Student — Capabilities: AI tutor, Voice learning, Simulations, Quizzes, Projects, Certifications, Learning memory, Collaboration. Teacher — Capabilities: Monitor students, View strengths, View weaknesses, Create learning groups, Generate quizzes, Generate assessments, Track progress. Admin — Capabilities: Manage platform, Manage users, Deploy updates, Manage schools, Manage system settings.

NAVIGATION STRUCTURE

Home; Learn; Practice; Create; Projects; Achievements; Certificates; Collaborate; Teacher Portal; Settings

ADAPTIVE HOME SCREEN

First-Time Learner Display: Welcome, Ask Anything, Explore Topics. Returning Learner Display: Continue Learning, Recent Projects, Achievements, Recommendations, Suggested Practice, AI Mentor.

LEARNING MODES

Learn Mode — Purpose: Understanding. Examples: What is gravity? What is entrepreneurship? Outcome: Understanding. Practice Mode — Purpose: Reinforcement. Includes: Exercises, Questions, Challenges. Outcome: Retention. Apply Mode — Purpose: Real-world use. Examples: Budgeting, Programming, Agriculture, Business. Outcome: Application. Create Mode — Purpose: Build something. Examples: Business Plans, Apps, Science Projects, Research Projects. Outcome: Creation. Teach Mode — Purpose: Teach OTIC. Student explains. OTIC evaluates. Outcome: Mastery.

AI TUTOR BEHAVIOR

OTIC never stops at answering. Response Pattern: Answer → Clarify → Practice → Apply → Create → Reflect.

LEARNING PATHS

When a learner asks: "Teach me Python" OTIC creates: Introduction, Variables, Conditions, Loops, Functions, Files, Projects, Assessment, Certification. This applies to all disciplines.

GAMIFICATION SYSTEM

Features: Points, Achievements, Badges, Levels, Milestones, Learning Streaks, Skill Trees. Examples: Science Explorer, Science Builder, Science Creator, Science Mentor.

CERTIFICATION SYSTEM

Upon successful completion generate: Skill Badge, Certificate, Completion Record, Stored Offline, Printable PDF.

COLLABORATION SYSTEM

Students may: Join Learning Groups, Collaborate on Projects, Share Ideas, Peer Learning, Challenge Friends, Compete Respectfully, Teach Each Other. Collaboration must remain educational.

SIMULATION ENGINE

Domains Supported: Science, Mathematics, Programming, Engineering, Business, Agriculture, Economics, Financial Literacy, Entrepreneurship, Health Education. Simulation Flow: Student Request → OTIC Suggests Simulation → Student Approves → Simulation Launches.

VOICE LEARNING

Offline Speech-To-Text, Offline Text-To-Speech, Voice Commands, Voice Tutoring, Voice Practice. Store: Text. Never Store: Audio Recordings.

STUDENT MEMORY ENGINE

Store: Age, Interests, Learning Style, Strengths, Weaknesses, Projects, Achievements, Certificates, Progress, Goals. Store summaries only. Never store unnecessary conversations.

TEACHER DASHBOARD

Overview, Students, Performance, Achievements, Weak Areas, Strength Areas, Projects, Assessments, Recommendations. Teachers can: Guide, Advise, Support, Mentor.

ADMIN DASHBOARD

User Management, Role Management, Update Management, Device Management, School Management, Analytics, System Monitoring.

EMOTIONAL SAFETY ENGINE

Detect: Bullying, Harassment, Abuse, Isolation, Anxiety, Stress, Depression Indicators. Response Strategy: Listen, Support, Recommend Trusted Adult, Recommend Teacher, Recommend Parent, Recommend Counselor. Critical Cases: Display Immediate Support Guidance.

KNOWLEDGE DOMAINS

Academic: Mathematics, Physics, Chemistry, Biology, Geography, History, English. Technology: Programming, AI, Data Science, Robotics. Business: Entrepreneurship, Marketing, Finance, Sales. Agriculture: Farming, Livestock,

Agribusiness. Life Skills: Communication, Leadership, Innovation, Critical Thinking. Professional Skills: Project Management, Digital Skills, Research Skills.

OFFLINE REQUIREMENTS

Internet Required: NO Cloud Required: NO External APIs: NO

UPDATE STRATEGY

Method 1: USB Flash Drive. Method 2: Local School Server. Purpose: Scalable deployment without internet.

DEVICE REQUIREMENTS

Android Minimum: 4 GB RAM, 32 GB Storage. Windows Minimum: 8 GB RAM. Ubuntu Minimum: 8 GB RAM.

AI MODEL STRATEGY

Phase 1: Gemma 3 1B. Benefits: Fast, Lightweight, Offline Friendly. Phase 2: Gemma 4. Optional Upgrade. Downloaded Separately.

SUCCESS METRICS

A learner should: Understand Concepts, Apply Knowledge, Build Projects, Solve Problems, Create Value, Teach Others, Collaborate Effectively, Earn Certifications, Develop Real Skills.

ULTIMATE GOAL

OTIC Studio should become: An AI Tutor + An Offline School + A Learning Operating System + An Education Platform that empowers learners to become creators, innovators, entrepreneurs, professionals, and lifelong teachers.